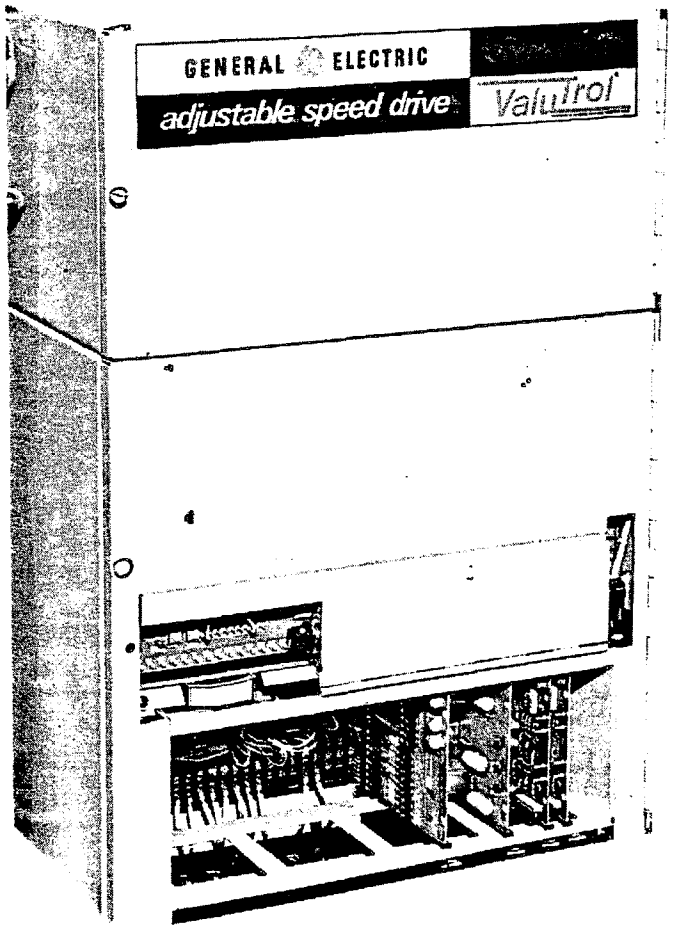
Preliminary GEK-85777

INSTRUCTIONS

ER2000

INCORPORATING VALUTROL®
MAIN CONTROL MODULE (MCM4)

INSTALLATION — OPERATION — MAINTENANCE



These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

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TABLE OF CONTENTS

Page

GENERAL DESCRIPTION 4
- Power Supply Card 4
- Main Control Card 4
- Auxiliary Control Card 4
- Test Instrument and Probe 6
- Interface Card 8
- Control Power 9
- Synchronizing Signal Power 9
- Control Connections 9
- Modification Rack 9

START—UP 10

SEQUENCE OF OPERATION 16
- Power Applied 16
- Switch Logic 16
- Signal Flow 16
- Stop 16

TROUBLESHOOTING 21
- Recommended Instrumentation 21
- Procedures 21

REMOVAL/REPAIR 22
- Printed Circuit Cards 22

GLOSSARY OF TERMS 23

HOT LINE TELEPHONE NUMBER ..... Back Page

LIST OF ILLUSTRATIONS

Figure		Page
1	Typical VALUTROL Main Control Module Block Diagram	5
2	Power Supply Card	6
3	Auxiliary Control Card	6
4	Main Control Card	7
5	Interface Card	8
8	Diagnostic Card and Modification Rack	9
9	Component and Signal Connection Location	11
10	Typical Test Data Sheet	15
11	Current Feedback (CFB) at Low Current Level	17
12	Current Feedback (CFB) at Continuous Current Level	17
13	Voltage Feedback (VFB) at Low Current	17
14	Voltage Feedback (VFB) at Continuous Current	17
15	Oscillator (OSC)	18
16	Initial Pulse (IPU)	18
17	Synchronizing Signal (SA)	18
18	Speed Feedback Signal (SFB) with AC Tach 450 RPM	18
19	Tachometer Feedback Signal (TFB) with AC Tach 450 RPM	19
19	Tachometer Feedback Signal (TFB) with AC Tach 3160 RPM	19
21	Speed Feedback Signal (SFB) with AC Tach 3160 RPM	19
22	Pulse Output (PO) Normal	19

LIST OF TABLES

I	Signal Connections	13
II	Fault Conditions	14

GENERAL DESCRIPTION

The basic elements of a typical **ER2000** drive system are shown in the simplified typical block diagram, Figure 1. This diagram indicates the basic elements contained in the **VALUTROL** main control module which functions to control output from a three-phase full-wave conversion assembly.

Three-phase AC power enters the system through fuses and is fed into power conversion modules (SCR) where it is converted to adjustable voltage DC power for the DC motor armature. Two conversion modules may be used as in Figure 1 to provide motoring and regenerative control of the DC load in both forward and reverse directions or a single module may be used to provide one-half or one-quarter of the two module control. Normally, the system may have an AC contactor (MA) that removes power from the drive when it is stopped leaving the drive system otherwise completely energized.

The speed of the motor is proportional to the DC voltage applied to its armature. Speed is measured by motor CEMF (Armature voltage feedback with IR compensation). As an optional feature, speed can be measured by a tachometer generator directly connected to the DC motor.

The basic control is manufacured on four (4) removable printed circuit boards. These are the power supply card (PSC) the main control card (MCC) and the auxiliary control card (ACC) and the interface card (IFC). Other cards, including a diagnostic card are also available as optional modifications.

The three-phase input also supplies power to the control power transformer. This transformer is fitted with a 460/230V reconnectable primary winding and two isolated secondary windings: (1) 115V to operate main control module relays, (2) the second winding is a 50 volt center tapped secondary which provides AC input to the power supply card.

POWER SUPPLY CARD (PSC)

The power supply card rectifies the AC input and provides regulated plus and minus 20 volts for the printed circuit cards. Unregulated plus and minus 30 volts DC is also provided to drive the static logic switches and the MAX relay. All of the DC outputs are fused to protect the power supply card against overloads. The regulated plus and minus 20V DC outputs are protected against over voltage conditions caused by a power supply card failure.

MAIN CONTROL CARD (MCC)

The primary purpose of the main control card is to drive the conversion modules (SCR A) as commanded by the speed reference and feedback signals. Connection to the SCR gates is by way of the auxiliary control card (ACC).

This card also performs several additonal functions such as linear timing of the reference; current limit; "Ready to Run" indicator and various scaling and trimming adjustments.

A total of (12) potentiometers are provided on this card, ten (10) of which are accessible from the front of the controller. The eleventh potentiometer, the card zero adjustment **ZERO ADJ.**, is preset at the factory and should normally not be disturbed. The twelfth potentiometer is the line impedance compenstion adjustment, **LINE**. The ten accessible potentiometers are:

DAMP	**MAX SPEED**	**MIN SPEED**
CUR LIMIT	**GAIN**	**REF SCALE**
CEMF LIMIT	**RESPONSE**	**LIN TIME**
COMP		

When the drive is first placed into operation the actual top speed may be different from what might be expected due to minor variations between tachometers. By adjusting the **MAX SPEED** potentiometer, any variations between tachometers can be eliminated without disturbing any other adjustments in the drive.

AUXILIARY CONTROL CARD (ACC)

The primary function of this card is to combine phase control signals, leg current signals, and oscillator signals and amplify the resulting pulse trains and direct them to the appropriate SCR gates by way of twelve (12) pulse transformers. One secondary function is to scale the output of three (3) current transformers (CT's) in the three-phase AC power inputs to the conversion modules (SCR A).

Another function is to detect circulating overcurrent faults which would not be detected by the shunt in the motor armature loop.

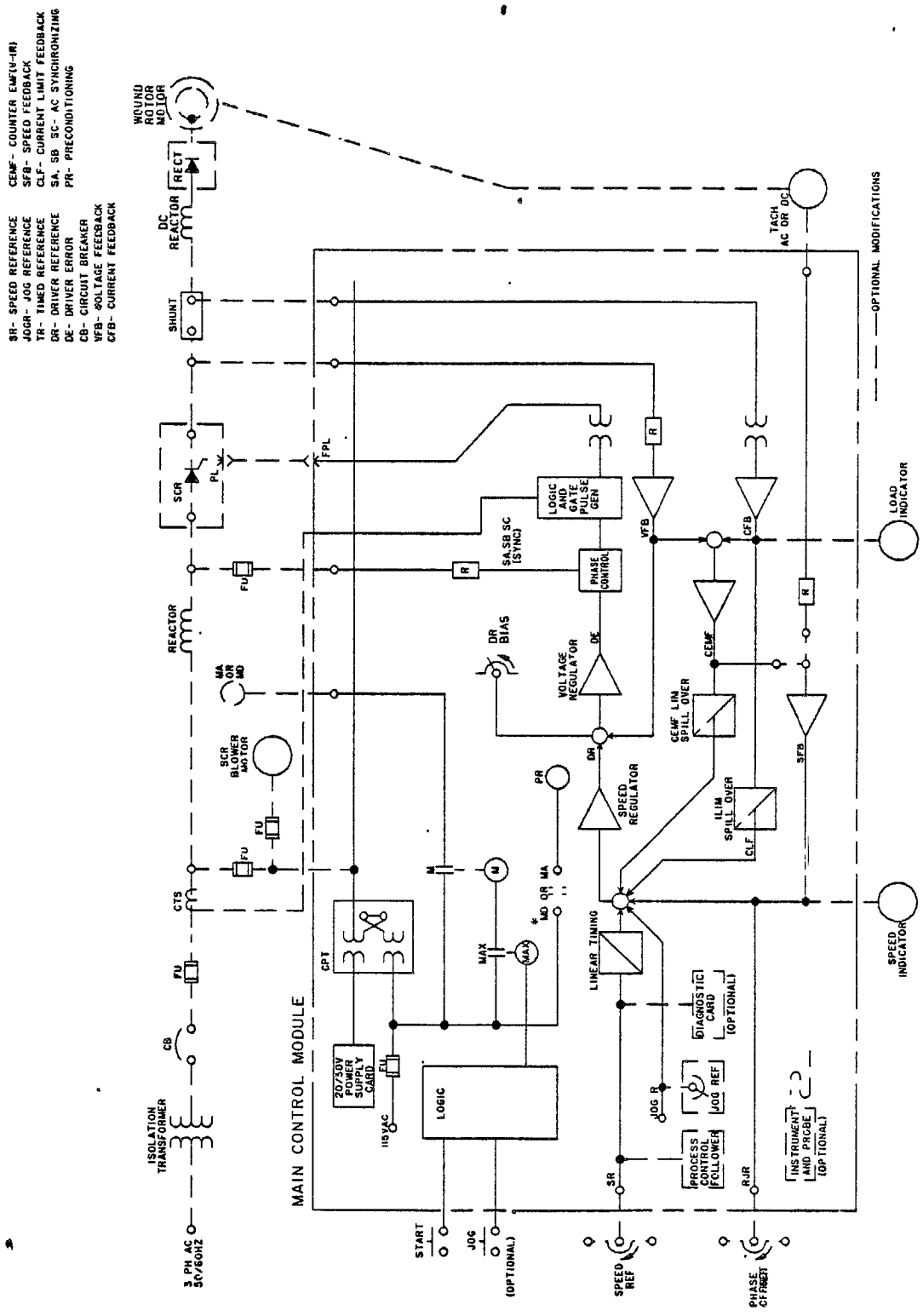


FIGURE 1
TYPICAL ER2000 DRIVE SYSTEM WITH VALUTROL®
MAIN CONTROL MODULE BLOCK DIAGRAM

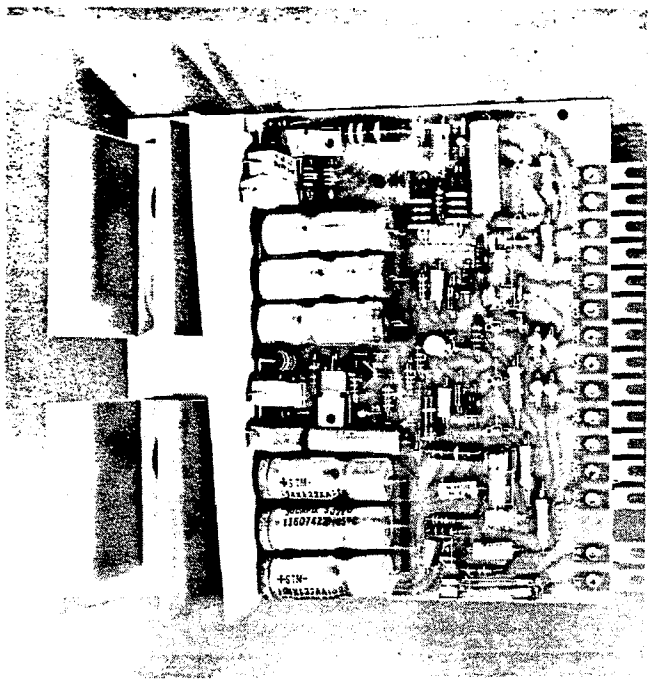


FIGURE 2
POWER SUPPLY CARD

An additional function is to generate a TRIP signal which suppresses SCR firing, executes a system fault, and may be used to operate an optional shunt trip circuit breaker.

TEST INSTRUMENT AND PROBE (OPTIONAL)
Located below the main control card (to the left) is a test instrument and probe that can be used to read out signals from any of the drive test points. The probe is fitted with two connections, one for the 4 volt instrument scale and other for the 20 volt scale. Always apply the 20 volt connection first. If the reading is below 4 volts, switch to the 4 volt connection for improved accuracy of the readout.

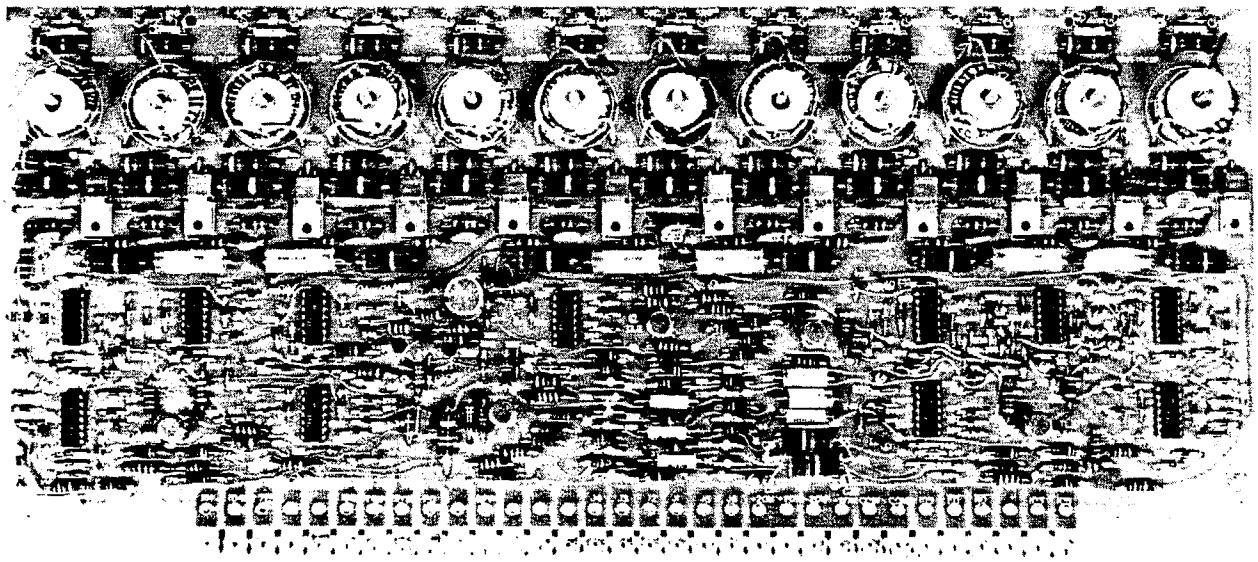


FIGURE 3
AUXILIARY CONTROL CARD

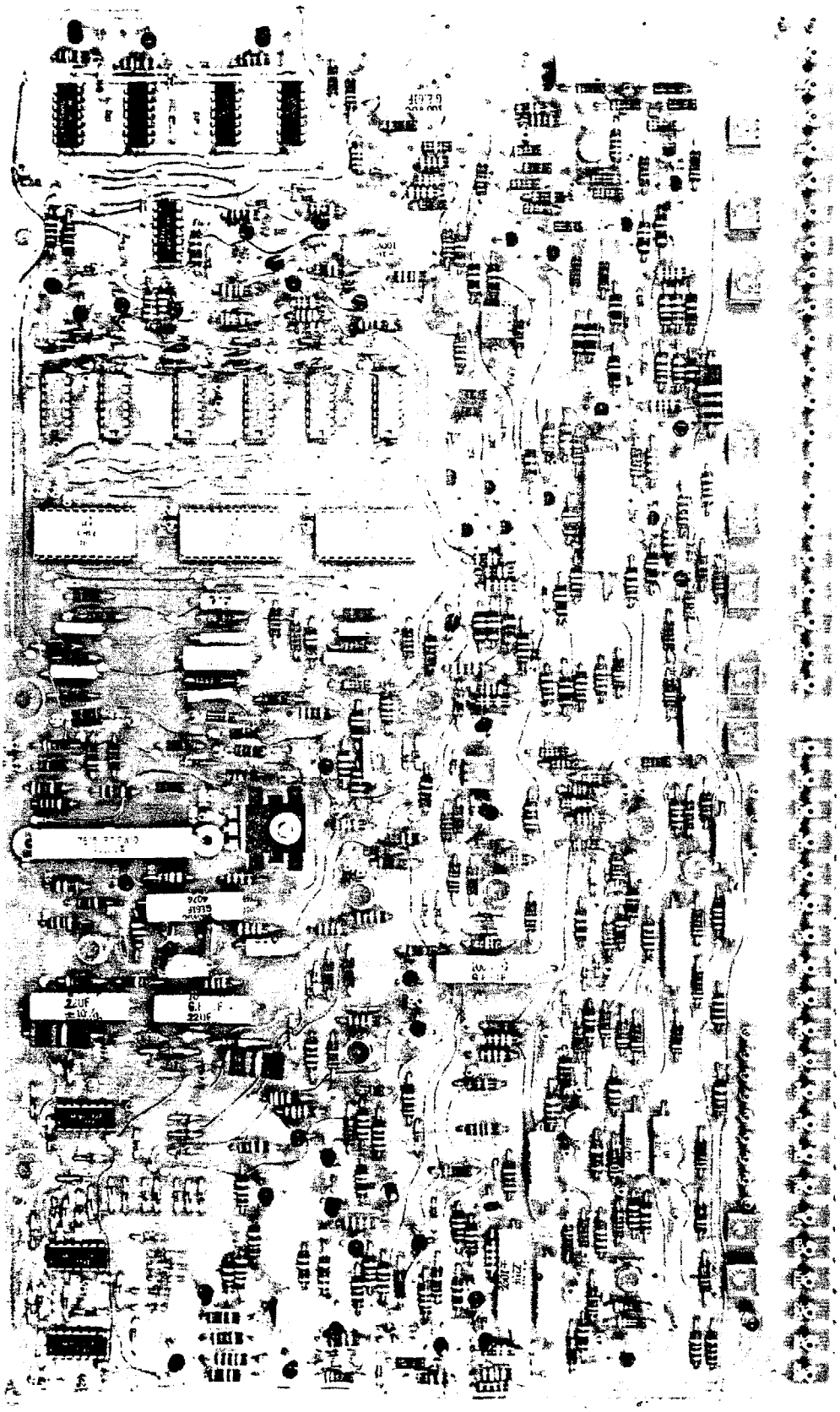


FIGURE 4
MAIN CONTROL CARD
(NOTE: TEST POSTS ALONG LARGE EDGE)

INTERFACE CARD (IFC)

The primary purposes of the interface card are:

1. To provide low level isolated signals corresponding to the three phase AC, DC armature voltage, armature and tachometer feedback.

2. To control the start, stop and synchronizing of the drive while monitoring the system for abnormal operation conditions.

3. To provide one milliampere signals for external speed and current indication.

Other outputs provide;

1. A NO/NC contact indicating M closure (MAX)

2. A NO contact indicating a fault condition (FLT).

   For those drives employing an AC tachometer an output whose frequency is proportional to RPM is generated which may be used to drive a digital counter. AC tachometer furnished by the factory will generate 18 pulses for each revolution.

There are (4) potentiometers on this card.

1. The **IZERO** is a bias adjustment for the current feedback output and is factory set. This control should not be disturbed.

2. **IMET** is the calibration adjustment for the current indicator.

3. **SMET** is the calibration adjustment for the speed indicator.

4. **RSTOP** is preset at the factory and should normally not be disturbed.

Adjustment 2 and 3 will be factory set if the indicators are ordered with the drive and mounted in the power unit enclosure.

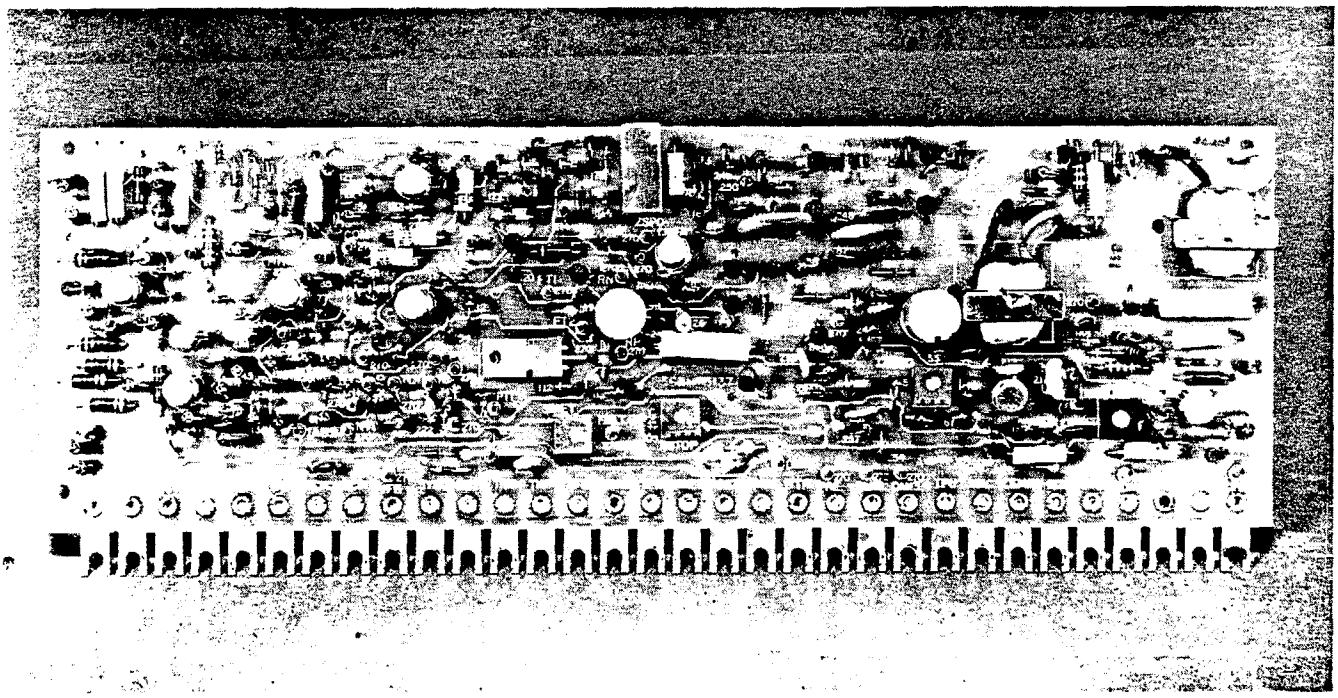


FIGURE 5
INTERFACE CARD

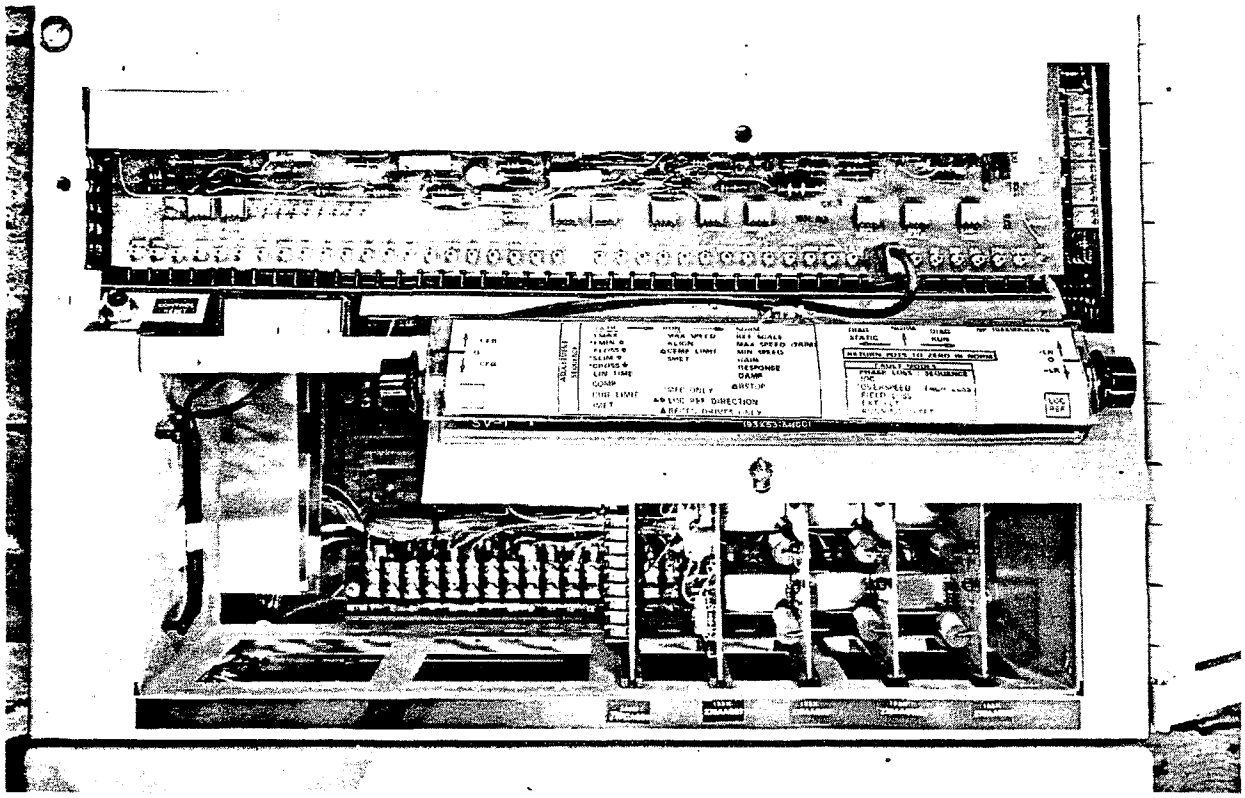


FIGURE 8
DIAGNOSTIC CARD AND MODIFICATION RACK
NOTE: DIAGNOSTIC CARD IS NOT FURNISHED

CONTROL POWER

Three phase power from an externally protected source is supplied to terminal board (TBN). The control transformer is connected to this power source.

SYNCHRONIZING SIGNAL POWER

Three resistance wires which provide line synchronization are connected through terminal board (TBN) to the AC line supplying the conversion module (SCR). Normally, these lines are fused in one of the conversion modules. These fuses also protect metal oxide varistors (MOV's) in the conversion module which protect that unit from excessive transient over voltage conditions. The drive will not operate if a fuse to a synchronizing lead is open.

CONTROL CONNECTIONS

Control connections are made to terminal boards 2TB, TBC and TBN and through plug FPL. Signals appearing on these terminal boards and their functions are described in Table 1. Optional connections are made at 3TB and 4TB. Refer to system elementary diagram for details. The SCR firing signals and each conversion module's over temperature thermal switch signal are carried through the plugs.

MODIFICATION RACK (MDR) (OPTIONAL)

Special features or functions which are related to the operation of the drive such as:

Special reference
Up to speed
Independent timed acceleration and deceleration adjustments
Etc.

are provided in printed circuit cards in the modification rack located below the main control card.

START—UP

The **VALUTROL** main control module is factory tested with the complete drive system. It is ready to operate provided the external power and control connections have been properly made and the following step by step procedures are followed:

1. Verify that the terminal board screws are tight.

2. Verify that incoming power is the proper voltage and the incoming wiring is complete and correct.

3. Apply power to the drive. If the green "Ready to Run" light located on the lower left hand corner of the main control card (MCC) is not illuminated, press and release the RESET pushbutton on the panel below. If the light does not turn on, the most probable cause is incorrect incoming phase rotation. Remove power, reverse any two of the incoming AC power leads and repeat.

4. Run the drive from the external speed reference up to top speed. Adjust **MAX SPEED** as may be required. DO NOT OVERSPEED.

5. Close and secure the doors of the power unit.

NOTE

ALWAYS READ THE COMPLETE INSTRUCTIONS PRIOR TO APPLYING POWER OR TROUBLESHOOTING THE EQUIPMENT. FOLLOW THE START UP PROCEDURE STEP BY STEP.

READ AND HEED ALL WARNING, CAUTION, AND NOTE LABELS POSTED ON THE EQUIPMENT.

WARNING: HIGH VOLTAGE

ELECTRIC SHOCK CAN CAUSE PERSONAL INJURY OR LOSS OF LIFE. WHETHER THE AC VOLTAGE SUPPLY IS GROUNDED OR NOT, HIGH VOLTAGE TO GROUND WILL BE PRESENT AT MANY POINTS. WHEN INSTRUMENTS SUCH AS OSCILLOSCOPES ARE USED TO WORK ON LIVE EQUIPMENT, GREAT CAUTION MUST BE USED. WHEN ONE OF THE INSTRUMENT LEADS IS CONNECTED TO THE CASE OR OTHER METAL PARTS OF THE INSTRUMENT, THIS LEAD SHOULD NOT BE CONNECTED TO AN UNGROUNDED PART OF THE SYSTEM UNLESS THE INSTRUMENT IS ISOLATED FROM GROUND AND ITS METAL PARTS TREATED AS LIVE EQUIPMENT. USE OF AN INSTRUMENT HAVING BOTH LEADS ISOLATED FROM THE CASE PERMIT GROUNDING OF THE CASE, EVEN WHEN MEASUREMENTS MUST BE MADE BETWEEN TWO LIVE PARTS.

CAUTION

DO NOT REMOVE PRINTED CIRCUIT CARDS FROM THE EQUIPMENT WHILE POWER IS APPLIED. THIS CAN DAMAGE THE EQUIPMENT.

1. FORWARD & REVERSE FIRING CABLE PLUGS (FPL & RPL)
2. TBN
3. TBC
4. CONTROL POWER TRANSFORMER (CPT)
5. PERMISSIVE RELAY (PR)
6. M CONTACTOR (M)
7. MTB
8. FAN
9. INTERFACE CARD (IFC)
10. POWER SUPPLY CARD (PSC)
11. MA AUXILIARY RELAY (MAX)
12. 2TB
13. FAULT RELAY (FLT)
14. 3TB
15. 4TB
16. MODIFICATION RACK (MDR)
17. S-22 CARDS
18. MAIN CONTROL CARD (MCC)
19. AUXILIARY CONTROL CARD (ACC)

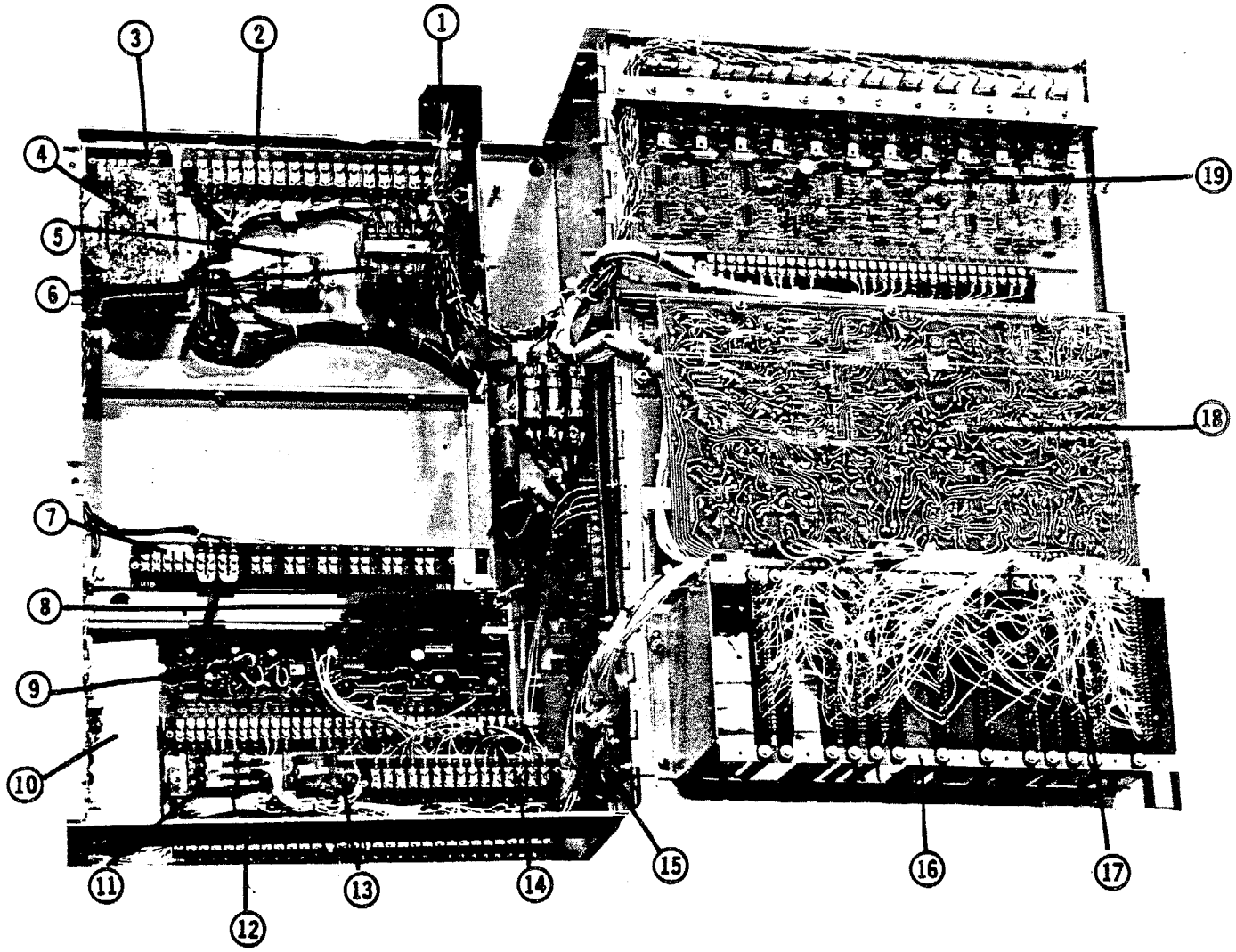


FIGURE 9
COMPONENT AND SIGNAL CONNECTION LOCATION

TABLE I SIGNAL CONNECTIONS

2TB	NOMENCLATURE	DESCRIPTION
1	-30V	Unregulated negative DC voltage used as the return line for the CONTROL ON function and the static switches RUN and JOG, AND possible modifications.
2	CONTROL ON	If CONTROL ON is not connected to —30V the drive will not start. If CONTROL ON is opened with the drive operating, the contactor will open and the drive will coast.
3,4	FLT	A normally open, held closed relay contact. Under normal conditions this contact is closed. If a fault condition is detected, this contact opens.
5,6,7	MAX	A NO/NC relay contact which actuates when the contactor actuates.
8,27	COM	Signal common. All signals are measured with respect to common, unless otherwise noted.
9	EST	External Fault Stop input. If EST is momentarily disconnected from common, the contact M Relay will open and the motor will coast. The drive may not be restarted until the reset line is momentarily connected to COMMON (2TB—12). (See RESET below)
10,11	FX1,X2	The internal 115V AC. FX1 is fused for external use.
12	RSET	Reset input. All fault shut downs inhibit the drive from starting until the fault has been cleared and the drive is reset. After the motor has come to a stop, the drive may be reset momentarily connecting RSET to common. The drive will not restart until RSET is released from common. Momentarily connecting RSET to common or pushing the RESET BUTTON will initiate a coast stop shutdown.
13,17 18,19 24,25	SP1, SP4 to SP8	These are special purpose wires which are used to bring additional signals out of 2TB. Refer to the system elementary for details. Additional SP wires may be connected to 3TB and 4TB as required.
14	RUN	The drive will not start unless either RUN or JOG are connected to —30V, either at 2TB or by special purpose logic in the MDR. When RUN and JOG are released from —30V, the M Relay is denergized and the drive will coast stop.
15,16	RPL3, FPL7	Connections to conversion module thermo-switches When more than one thermo-switch exists, they will be connected in series.

TABLE I SIGNAL CONNECTIONS

2TB NO.	NOMENCLATURE	DESCRIPTION
20,21	+20V, -20V	Regulated power supply outputs.
22	IMET	Output to an optional 1ma load instrument. The instrument is calibrated with the **IMET** potentiometer on the Interface card.
23	SMET	Output to a 1ma speed instrument. The instrument is calibrated with the **SMET** potentiometer on the Interface Card.
26	SMIN	output from the **MIN SPEED** potentiometer on the main control card.
28	SR	Speed Reference input.
29,30	TKP, TKN	Input connections for motor mounted tachometer or machine mounted tachometer. NOTE: WITH A DC TACHOMETER, TKP MUST BE POSITIVE WHEN SYSTEM REFERENCE IS NEGATIVE AND DA1 IS POSITIVE WITH RESPECT TO DA2.

TBA NO.	NOMENCLATURE	DESCRIPTION
J1 thru J6	J1 thru J6	Current transformer feedback from AC line to ACC for detection of circulating current.

TBC NO.	NOMENCLATURE	DESCRIPTION
1	PS-2X (X2)	Internal 115VAC (grounded to case).
2	PS—3	Internal 115VAC applies through a NO "M" relay contact.
3,5	M	NC "M" Relay contacts between TBC(3) and TBC(5) signal relay closure.
4	PR	PR Relay coil. Connect TBC(4) to TBC(2), energizes PR Relay making firing cirucit operative.

TBN NO.	NOMENCLATURE	DESCRIPTION
K1,K2,K3	EXC—P	Single phase power for control transformer (CPT).
T11,T12,T13	SYNC SIG	Synchronizing signals from converter power source.
DC1, A2	V FDBK	Voltage Feedback from DC motor armature
1P,1N	CURR FDBK	Current Feedback signal from DC line, 100MV shunt.
F1,F2	MTR FLD	Not used

TABLE II FAULT CONDITIONS

A fault has occurred if the fault relay contact (FLT) is open or if the "READY TO RUN" light is off. The conditions that can initiate a fault are as follows:

1. No three phase power.
2. Loss of incoming phase.
3. Incorrect phase rotation.
4. AC power fuse blown.
5. Control fuse is open.
6. Power supply plus or minus DC fuse is open.
7. Instantaneous overcurrent (IOC) level exceeded.
8. Motor thermo-switch (OLD). (If connected to fault circuitry).
9. Timed overcrrent (TOC) - electronic. (If connected to fault circuitry).
10. External Fault Stop momentarily released from Common.
11. Other special functions to System Trip (SYS) or External Fault Stop inputs.
12. System Trip input (SYS) momentarily connected to +10 volts.
13. RESET button depressed or RSET input momentarily connected to Common with motor rotating.
14. RESET button held depressed or RSET input held connected to Common.
15. Oscillator (OSC) failed "on".
16. Trip actuated from ACC (by AC overcurrent or SCR commutation failure).

After fault condition has been cleared and the motor has come to a standstill, the drive can be RESET by any of the following three methods:

#1. Momentarily remove the three-phase power and re-apply.

#2. Push the RESET button.

#3. Momentarily connect RSET to common.

If all fault conditions have been cleared but the drive fails to RESET, the RSTOP adjustment may be set too low.

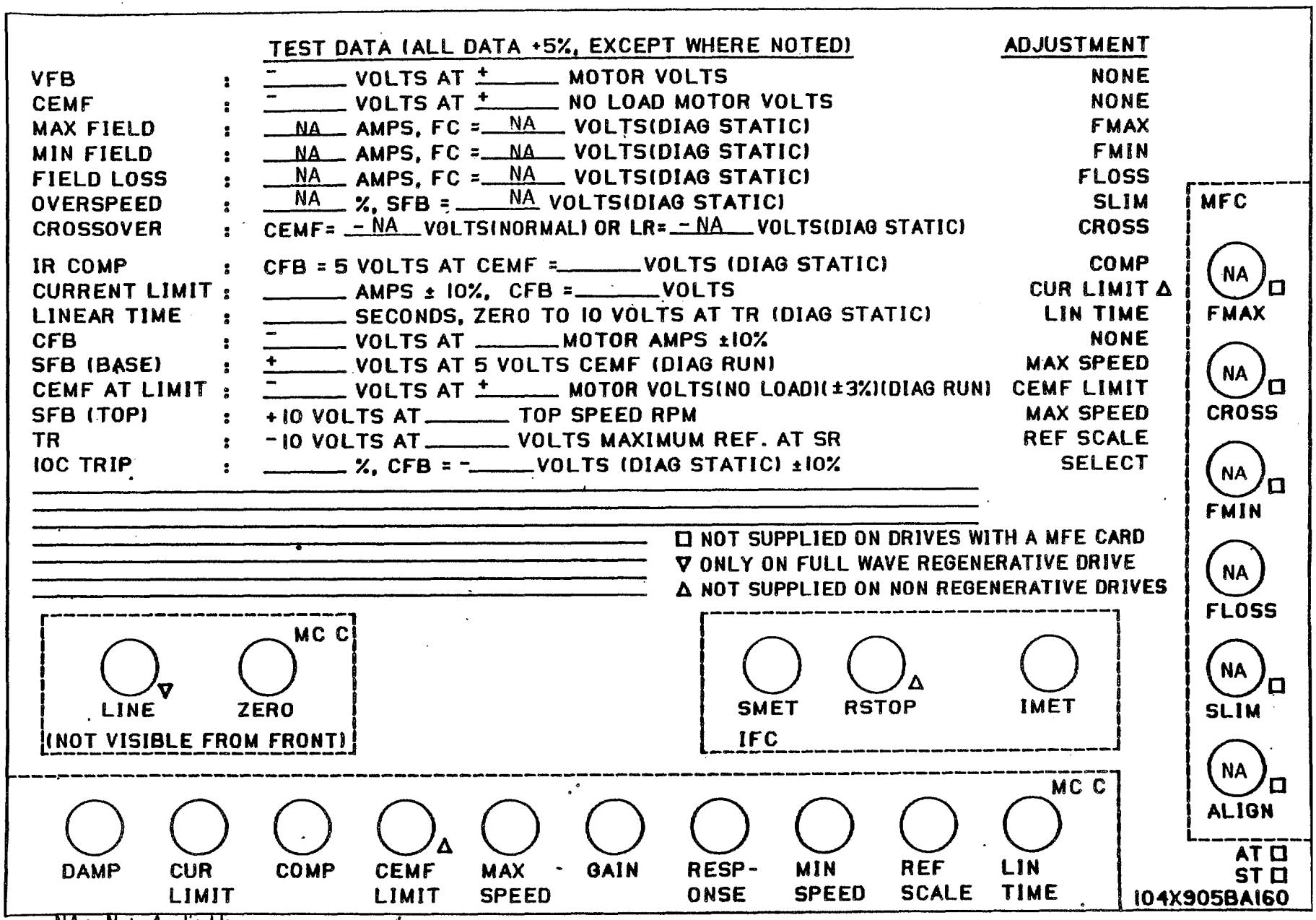


FIGURE 10
TEST DATA SHEET

NA: Not Applicable

SEQUENCE OF OPERATION

POWER APPLIED

The control transformer (CPT) is energized through its primary fuses.

The power supply card is energized and the DC output (±20 volt) are applied through their fuses to the rest of the cards. All readings carry a tolerance of ±10%.

If no faults have been detected by the monitor section of the interface card, the fault relay FLT will close, and the "Ready to Run" indicator on the main control card will illuminate. Table II tabulates the fault conditions which are monitored.

The oscillator will start, and the synchronizing signals SA, SB, SC will measure 8.5 volts RMS, (±10%). See Figure 17.

SWITCH LOGIC

RUN or JOG will be switched from +30V to —30 volts. (under purchaser control).

The control line MAC from the main control card to the interface will be pulled down to —20 volts.

The interface card checks that no faults exist and that "control on" is connected to —30 volts before applying power to the coil of the pilot relay MAX.

MAX picks up, releasing the preconditioning signal PRE from common and applies power to the coil of the M relay which picks up the contactor.

When PRE is released from common, it switches to —4 volts which will release the main control card preconditioning after approximately 80 milliseconds.

Releasing preconditioning allows the drive to send firing pulses to the gates of the SCR's in the conversion module, and allows the normal flow to occur.

SIGNAL FLOW

If RUN is switched, the reference at SR is applied to linear time section. The timed reference output TR will ramp to a voltage proportional to SR. The **REF SCALE** adjustment is used to set TR to 10.0 volts when the input at SR is set for top speed. The time for TR to ramp from 0 to 10 volts is adjustable from .3 to 60 seconds with **LIN TIME** adjustment. See jumper table on system elementary. (Ranges: .3 to 10 sec. or 2 to 60 sec.)

The speed (or CEMF) feedback from the motor tachometer is scaled with a selectable resistor network on the interface card, and rectified (if required) on the main control card.

The output of the speed feedback section is SFB, and will be 10 volts at top speed. **MAX SPEED** is adjusted to make the actual top speed correspond to desired top speed. See Figures 18 and 21.

The timed reference TR, the JOG reference JOGR, and the speed feedback SFB are summed by the regulator error amplifier. The error amplifier output EAO will be a low voltage (nearly zero) when the drive is regulating speed. EAO will not be low when the drive is current limit or CEMF limit. The gain of the error amplifier is set with the **GAIN** adjustment. The **GAIN** adjustment is used to improve the response of the drive.

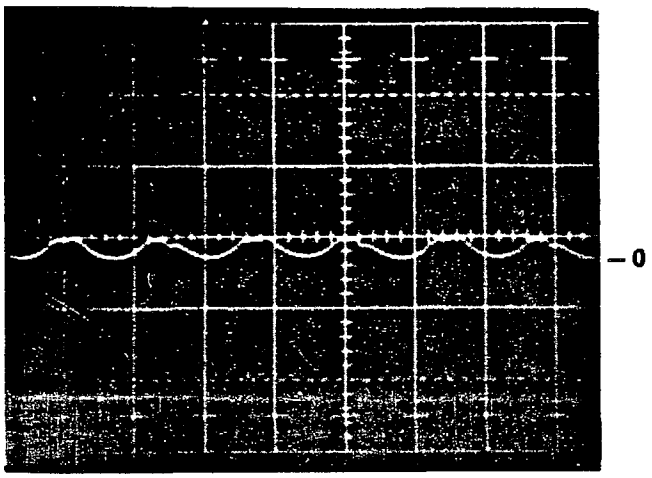
To maintain good load regulation, the error amplifier is fed into the regulator integrator. The output of the integrator is the reference, DR, to the driver. The response of the control is set with the **RESPONSE** adjustment.

There is a limit, however, to how responsive a drive may be set. Stability of the drive is decreased as its response is increased.

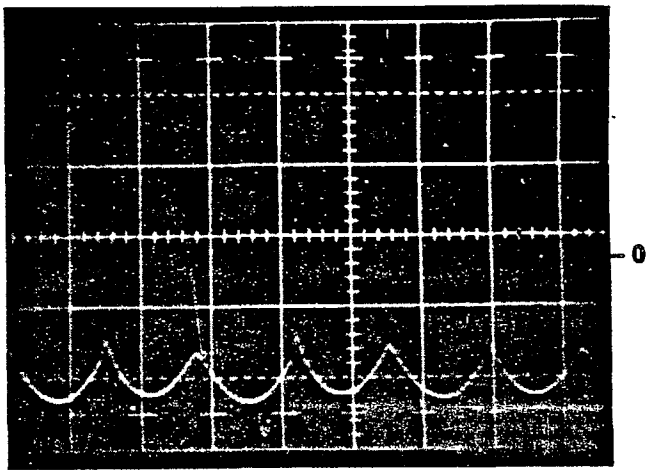
To protect the system, three limit sections are provided; counter—EMF (CEMF) limit and two current limits. The output of the CEMF limit and the primary current limit drive the regulator integrator and will overide the error amplifier, if required. The primary current limit is set with the **CUR LIMIT** adjustment and the counter EMF limit with the **CEMF LIMIT** adjustment. Typically, the primary or regulator current limit is set at 150% of the motor nameplate current of 3.75 volts (±10%) of current feedback, CFB. The counter EMF is normally limited to 250 armature volts at no load, or 5.75 volts (±10%) of CEMF for drives rated 240 volts DC. For drives rated 500 volts DC the CEMF limit is normally 510 volts at no load or 560 volts (±10%) at CEMF. The secondary or driver current limit is in the driver section and will be described later. The primary current limit, if used, should not be set higher than the driver current limit. The driver current limit may be inhibited or adjusted.

WAVEFORMS

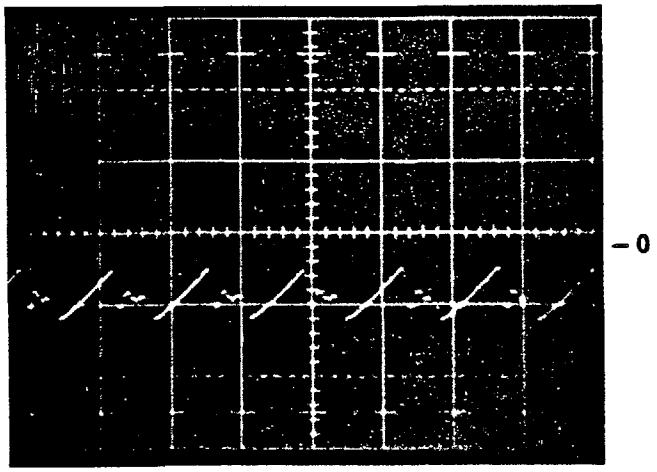
All illustrations were photographed in the forward motoring quadrant with zero volts on center line at 2 msec per division.



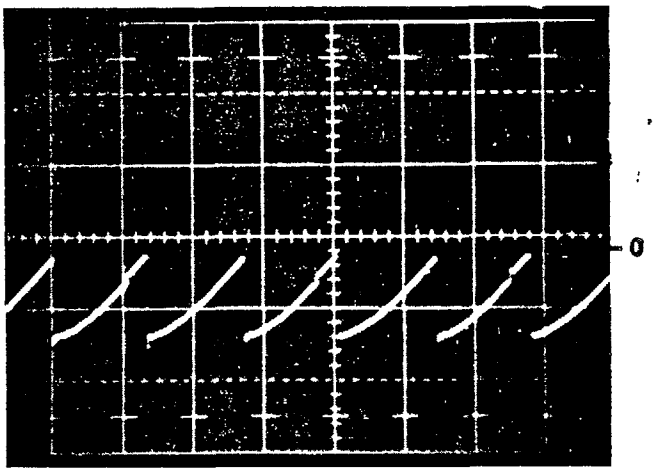
2 msec/div
At low current level 1 volt/division
FIG. 11 CURRENT FEEDBACK (CFB)



2 msec/div
At Continuous current 1 volt/division
FIG. 12 CURRENT FEEDBACK (CFB)



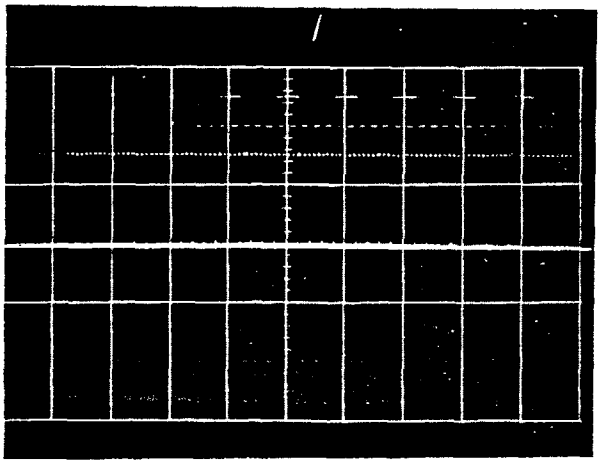
2 msec/div
At low current and 200 volts 5 volts/division
FIG. 13 VOLTAGE FEEDBACK (VFB)



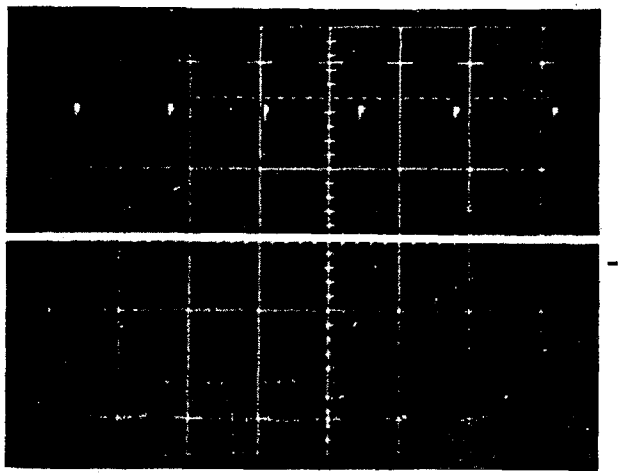
2 msec/div
At continuous current and 200 volts 5 volts/division
FIG. 14 VOLTAGE FEEDBACK (VFB)

WAVEFORMS

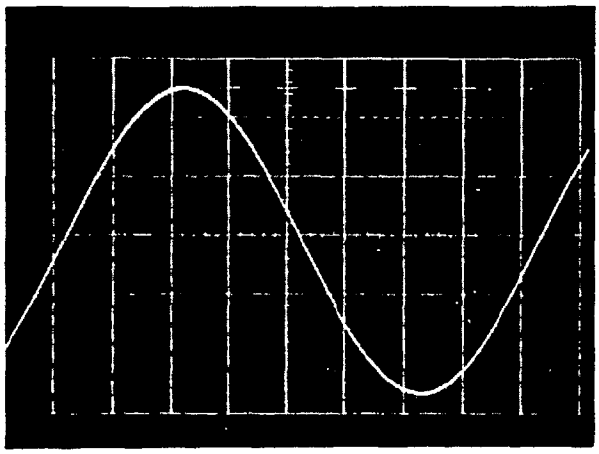
All illustrations were photographed in the forward motoring quadrant with zero volts on center line at 2 msec per division.



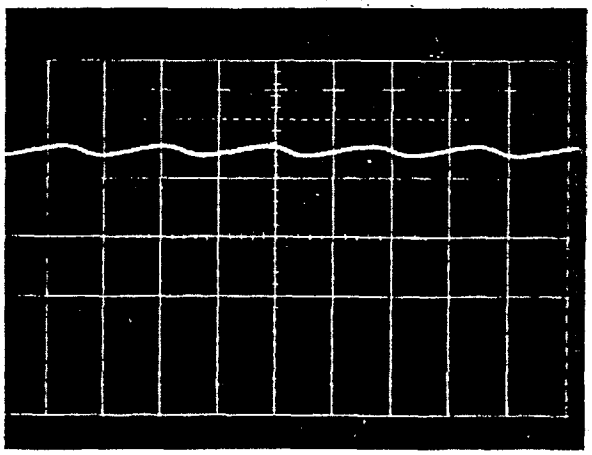
2 msec/div
10 volts/division
FIG. 15 OSCILLATOR (OSC)



2 msec/div
2 volts/division
FIG. 16 INITIAL PULSE (IPU)



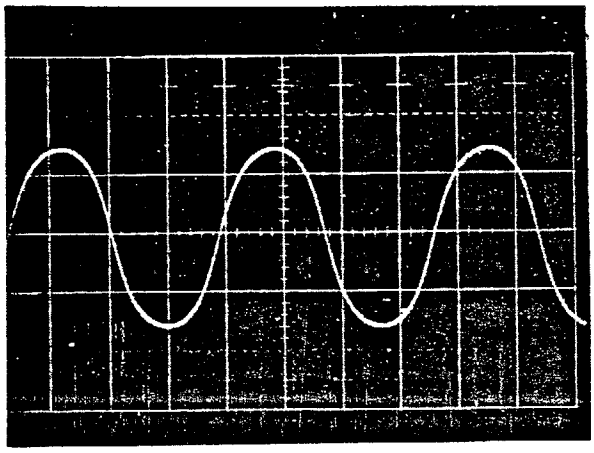
2 msec/div
Typical of SA, SB & SC
SB lags SA by 120°
SC lags SB by 120°
FIG. 17 SYNCHRONIZING SIGNAL (SA)



2 msec/div
With an AC tachometer at 450 RPM
1 volt/division
FIG. 18 SPEED FEEDBACK (SFB)

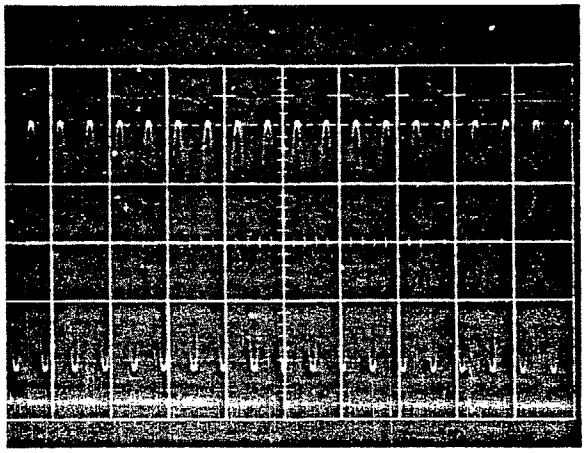
WAVEFORMS

All illustrations were photographed in the forward motoring quadrant with zero volts on center line at 2 msec per division.



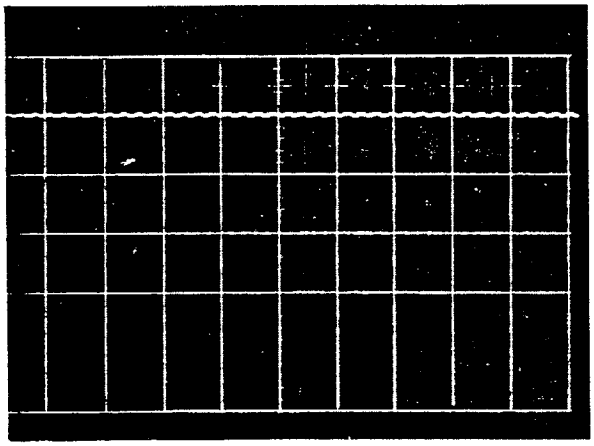
2 msec/div
With an AC Tachometer at 450 RPM
1 volt/division

FIG. 19 TACHOMETER FEEDBACK (TFB)



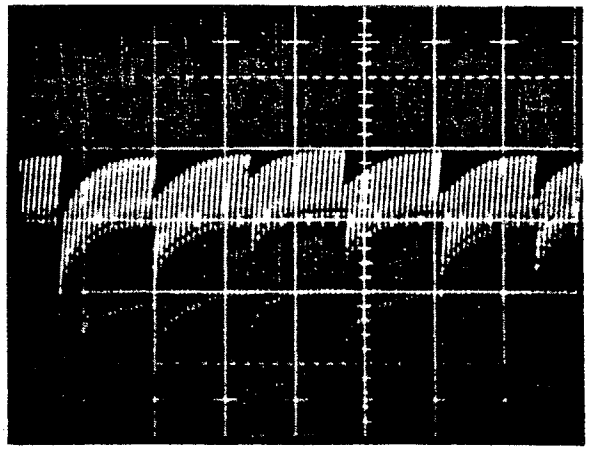
2 msec/div
With an AC Tachometer at 3160 RPM
5 volts/division

FIG. 20 TACHOMETER FEEDBACK (TFB)



2 msec/div
With an AC Tachometer at 3160 RPM
5 volts/division

FIG. 21 SPEED FEEDBACK (SFB)



2 msec/div
AC Coupled
0.1 volt/division

FIG. 22 PULSE OUTPUT (PO)

The main purpose of the secondary, or driver, current limit is to be an operating current limit if the regulator, or primary current limit, is not used and the system is controlled by other signals summed at the driver junction. The driver current limit may be increased by connecting a jumper from DCX to DCY on the main control card which will increase the limit approximately 130% to 175% of the motor nameplate current. Alternative levels may be established by connecting a resistor between DCL and DCX on the main control card. For normal levels of regulating current limit, either DCX should be jumpered on DCY to increase the driver current limit or it should be inhibited by connecting DCI to COM. For regulating current limit above 150%, a resistor should be connected between DCL and DCX to raise the driver limit above 175% or it should be inhibited with a jumper between DCI and Com.

The counter—EMF signal CEMF is developed on the main control card by subtracting a signal proportional to IR drop of the motor from voltage feedback. This is set with the COMP adjustment.

The driver reference, DR, the voltage feedback, VFB, an armature current signal from damping adjustment, **DAMP**, and the driver current limit output are summed at the input to the driver. The driver converts this error to pulse trains which drive the SCR rates in such a manner as to maintain the motor voltage proportional to the driver reference. The damping adjustment **DAMP** controls the response of the driver.

Generally speaking, **DAMP** is used to quiet small oscillations which occur in the current under light load conditions. Too much damping will slow down the system and response and tend to cause over shoot.

The driver provided a signal IPU to the oscillator on the interface card to generate an initial pulse at the exact point in time that an SCR is to be fired. See Figure 16.

Two driver monitor points are available, PCR and PO, PCR is the phase control reference which causes the output pulse trains to phase shift in time with respect to AC line. As PCR moves from zero to +6 volts (±10%), the output pulses will shift from full off to full on. PO is used to monitor the pulse outpouts to the SCR's. See Figure 22.

STOP

There are two stop sequences, normal stop and fault stop. With a normal stop the drive regenerates to near zero speed before opening the contactor. A fault stop opens the contactor and drive coasts to a stop.

Normal stop (disconnect RUN from —30 volts).

RUN will switch from —30 volts to +30 volts. MAC will switch to zero volts and the system reference input to the linear time section will be shunted to common.

The timed reference TR will begin to time down to zero and the drive speed will come down accordingly.

The regenerative stop circuit on the interface card will hold the contactor closed until the CEMF signal is almost zero, corresponding to zero speed. At this time, the preconditioning signal PRE goes to common, removing power from the MAX coil. 100 milliseconds later, MAX drops out removing power from M, which then drops out. The CEMF level corresponding to zero speed is set by the **RSTOP** adjustment. If **RSTOP** is set too far (CW), power is removed prematurely and the drive will coast into zero speed. If **RSTOP** is set too far (CCW) the contactor will not open at all.

In some cases the regenerative stop circuit, (described above) may be under the control of the speed feedback signal, SFB, rather than the CEMF signal.

FAULT STOP—Fault detected (See Table II) or CONTROL ON is open.

The preconditioning signal PRE is immediately supplied to the main control card, forcing the drive into zero current or coast conditions. As soon as the current goes to zero, preconditioning is established throughout the card.

The contactor unconditionally drops out 100 milliseconds after the fault condition occurs.

The drive can not be restarted until the motor has come to rest. If the STOP was initiated by a fault, this is taken care of automatically, but it is the purchaser's responsibility to **not** reclose "CONTROL ON" before the motor has come to rest. After the motor has stopped, push the RESET button.

REF SCALE/MAX SPEED (reference scale/max speed)

Turn **REF SCALE** full CCW. Start the drive and apply top speed reference to SR. Adjust the **REF SCALE** potentiometer until SFB is 10 volts (±10%). This normalizes the timed reference TR and speed feedback, SFB for 10 volts (±10%) at top speed.

Now measure motor RPM and adjust **MAX SPEED** (if necessary) until the actual RPM corresponds to the desired top speed. If actual top RPM was off by more than 5% reset **ALIGN** as detailed above.

GAIN, RESPONSE and DAMP (Stability adjustments)

1. Set **DAMP** potentiometer at 7 o'clock position.

2. Set the **GAIN** potentiometer at minimum (7 o'clock).

   Small incremental changes in the GAIN setting may be required when adjusting RESPONSE with the RUNNING. See Step 4.

   See motor nameplate under — Speed.

3. Set **LIN TIME** potentiometer at minimum (7 o'clock).

4. Set **RESPONSE** potentiometer at minimum (7 o'clock).

   When the drive is functioning properly in all other respects make small incremental step increases and decreases in speed. Observe current feedback (CFB) for bumping repeatedly before steady state speed is attained.

   Increase the **RESPONSE** setting (move CW) until bumpy current is observed. Then reduce the **RESPONSE** setting until no bumps (or only one) is observed. This is the maximum **RESPONSE** setting.

   In general settings below 10 o'clock will show signs of increasing sluggishness. Setting greater than 2 o'clock may how signs of hard or even continuous bumping. Full **RESPONSE** setting (5 o'clock) will usually trip the IOC.

5. Reset the **LIN TIME** to required setting.

REF SCALE (reference scale)

Turn **REF SCALE** full CCW. Start the drive and apply top speed reference to SR. Adjust the **REF SCALE** potentiometer until SFB is 10 volts (±10%). This normalizes the timed reference TR and speed feedback, SFB for 10 volts (±10%) at top speed.

MIN SPEED (minimum speed)

Reduce the system reference to minimum and start the drive. Adjust **MIN SPEED** as required to meet system minimum speed requirements. Refer to system elementary diagram for circuit details.

TROUBLESHOOTING

Although many of the problems which may arise can be effectively located with a multi-meter, an oscilloscope is a very powerful troubleshooting tool. The only requirements are that the selected scope have a DC input capability and a line synchronization mode. Caution should be exercised in measuring any point with a possible high potential with any instrument; however, particular care should be taken with an oscilloscope since the common clip is normally connected directly to the instrument case. If the grounded plug has not been defeated, it will cause a short circuit between the high potential point under test and ground.

RECOMMENDED INSTRUMENTATION

Simpson Multi-meter (or equivalent) 10,000 ohms/volt (or higher).

Hewlett-Packard or Tektronic (or equivalent). Dual Trace oscilloscope rated for operation from DC to 10 MHZ at 0.01V/MC with deflection factors to provide 0.01V/CM to 1300 peak to peak deflection when used with appropriate attentuator probes.

PROCEDURES

In troubleshooting this drive system the most appropriate place to start is to follow the SEQUENCE OF OPERATION (previously described) until a descripancy or fault is noted. This step by step procedure will determine which part, sub-assembly or printed circuit is causing the problem.

REMOVAL/REPAIR

PRINTED CIRCUIT CARDS

There should be no need to retune the drive after removal/repair of a conversion module, an SCR or any other removable sub-assembly unless, of course, an adjustment was inadvertently moved or disturbed. If a printed circuit card is replaced (other than the power supply card PSC):

1. Add stab-on jumpers to the replacement card just like the jumpers on the card that was replaced or as listed on the System Elementary Diagram "Programming" Table.

2. Add stab-on resistors and capacitors to the replacement card just like the components on the card that was replaced or as shown with values on the system elementary main control card (MCC) at stab-on terminals on terminals TR, RJ, SFB, NDE, CLI, CLJ, and LT2 or on the DM1, DM2 etc. Terminals on any printed circuit card.

3. Set the potentiometers on the replacement printed circuit card to the position as was set on the card that was replaced or the position shown on the test data sheet. Recheck the calibration procedures described.

4. Use caution when connecting or disconnecting stab-on connectors on the printed circuit cards to avoid breaking of the connector posts. Support the card if possible and use a pair of long nosed pliers to hold on to the connector crimp. Avoid pulling on wires when removing connectors.

GLOSSARY OF TERMS

ACC — Auxiliary Card
ALIGN — Tachometer Loss Align Adjustment

*CEMF — Counter EMF
CEMF LIMIT — Counter EMF Limit Adjustment
*COMM — Regulator Common
COMP — IR Compensation Adjustment
CPT — Control Power Transformer
*CFB — Current Feedback
CUR REF — Diagnostic Current Reference Potentiometer
CROSS — Crossover Adjustemnt
CUR LIMIT — Current Limit Adjustment

DAMP — Dampening Adjustment
*DM1 — DM8 Dummy Input/Output points
*DR — Driver Reference

*EAO — Error Amplifier Output
EST — External Fault Stop

FLT — Fault Relay

GAIN - Speed Loop Gain Adjustment

IFC — Interface Card
IMET — Current (Load) Instrument Output and Adjustment
IOC — Instantaneous Over Current
*IPU — Initial Pulse

*JOG — Jog Switch Input
*JOGR — Jog Reference

LIN TIME — Linear Timing Adjustment

MA — Line Contactor
*MAC — MAX Control Signal
MAX — Pilot Relay for M
MAX SPEED — Adjustment
MCC — Main Control Card
MD — DC Line Contactor
MDR — Modification Rack
MIN SPEED — Adjustment
MOV — Metal Oxide Varistor

*OSC — Oscillator

*PCR — Phase Control Reference
PO — Pulse Outputs
PR — Permissive Relay
*PRE — Preconditioning
PSC — Power Supply Card

REF SCALE — Adjustment
RESPONSE — Speed Loop Response Adjustment
RESET — Pushbutton
RSTOP — Regenerative Stop Adjustment
*RTR — "Ready to Run" Indicator
*RUN — Run Switch Input

*SA, SB, SC — Synchronizing Signals
*SFB - Speed Feedback
*SMET — Speed Instrument Output and Adjustment
*SMIN — Minimum Speed Reference Adjustment and Input
*SR — Speed Reference
*SYS — System Fault Trip
*TA — Tachometer Align Output
*TFB — Tachometer Feedback Signal
TKN — Negative Tachometer Input
TKP — Positive Tachometer Input
*TR — Timed Reference
TRIP — Trip Signal for Optional Circuit Breaker
*VFB — Voltage Feedback
*TEST Points Located on Door Front (See MCC Illust., Fig. 4)

HOT LINE TELEPHONE NUMBER

The Contract Warranty for **VALUTROL** drives is stated in General Electric Apparatus Handbook Section 105, Page 71.

The purpose of the following is to provide specific instructions to the Valutrol—Drive user regarding warranty adminstration and how to obtain assistance on out-of-warranty failures.

1. In the event of failure or misoperation during "in warranty" refer to the instruction book to identify the defective part or subassembly.

2. When the defective part has been identified (or for assistance in identification) call:

GENERAL ELECTRIC COMPANY
ERIE, PENNSYLVANIA
814—455-3219
(24 Hour Phone Service)

or

Contact the nearest

General Electric Installation and Service Engineering Office listed in your telephone directory. Before calling, list model and serial numbers of the power unit and DC motor.

GENERAL ELECTRIC COMPANY
SPEED VARIATOR PRODUCTS OPERATION
ERIE, PENNSYLVANIA 16531

GENERAL ELECTRIC